



EMBEDDED SMART DESK
23EXC409 MINI PROJECT I REPORT



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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

KARPAGAM COLLEGE OF ENGINEERING

COIMBATORE – 641 032

ANNA UNIVERSITY : CHENNAI 600 025

DECEMBER 2025



**KARPAGAM COLLEGE OF ENGINEERING
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BONAFIDE CERTIFICATE

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DECLARATION

We hereby declare that this Project report entitled “**SMART DESK**”submitted by us for the degree of **BE in Electronics and communication Engineering at Karpagam College of Engineering, Coimbatore** is the record of original work done by us under the guidance and supervision of **Dr.R.Vadivelu,Mtech.,Ph.D ASSISTANT PROFESSOR** at then Department of Electronics and communication Engineering,Karpagam College of Engineering, Coimbatore–641032 and has not formed the basis for the award of any degree, or diploma or titles in this institution or any other Institution of higher learning.

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ACKNOWLEDGEMENT

First and foremost praises and thanks to the almighty for her showers and blessings throughout our project work to complete it successfully.

We extend our gratitude to the Management of Karpagam College of Engineering, Coimbatore for the excellent infrastructure and support facilities to undergo the project work.

We are very grateful to **Dr. P. Karthigaikumar**, the Principal and **Dr.R.Sarankumar ME., Ph.D., Electronics and Communication** for provided the facilities, support and permission to carried out our project work at our esteemed institution.

We record our sincere gratitude to our Project Coordinator of **MR.R.Vadivelu,M.tech(Ph.D)** for giving inputs, encouragement for the continuous improvement during the progress and to complete this project work.

We would like to express our sincere gratitude to our Supervisor **Mr.R.Vadivelu,M.tech(Ph.D)** for the continuous support for our UG study, for his motivation and adequate guidance which helped us to achieve success in all our accomplishments and to complete this project work.

We also thank all the teaching faculty members and non-teaching Staff members of the department of **Electronics and Communication**, Karpagam College of Engineering, Coimbatore for their kindness and support.

I would like to thank **our parents, family members and friends** who sacrificed their time and energy to complete the project work successfully.

ABSTRACT

This project presents the design and development of an Intelligent Embedded Smart Desk aimed at enhancing students' study efficiency, health awareness, and parental communication. The system integrates multiple sensors and control modules to automate essential desk functions. An LDR sensor and IR sensor are used to detect ambient light levels and user presence, enabling automatic lighting control for energy efficiency. A buzzer module acts as a hydration and break reminder, alerting the user every 45 minutes to drink water and take short rest intervals, thus promoting healthy study habits. To ensure parental awareness, the system employs a GSM communication module that automatically sends text messages to the parent's mobile number, informing them when the student is studying and when hydration reminders are acknowledged. The entire system is controlled by an Arduino microcontroller, which processes sensor inputs and executes automation logic in real time. This Smart Desk not only creates a comfortable and energy-efficient workspace but also incorporates wellness and monitoring features, making it ideal for modern educational environments. The project demonstrates the practical application of embedded systems, automation, and IoT concepts in improving both productivity and personal health.

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1. INTRODUCTION

1.1 Introduction to the Project :

- The Embedded Smart Desk project is designed to automate and enhance the comfort, convenience, and efficiency of a workspace using embedded system technology.
- The system integrates multiple sensors and modules controlled by a 5V ESP32 microcontroller, which serves as the brain of the project.
- An IR sensor is used to detect the presence or movement of a person near the desk, enabling automatic activation of certain functions such as turning on the lamp. The LDR (Light Dependent Resistor) sensor monitors the surrounding light intensity, allowing the system to automatically switch the 12V table lamp on or off depending on the ambient lighting conditions.
- To ensure environmental awareness, a DHT11 sensor is incorporated to measure the temperature and humidity around the desk, providing real-time monitoring of environmental comfort. The relay module acts as an interface between the low-voltage control system and the high-voltage lamp, ensuring safe and efficient operation.
- For remote communication and control, a GSM module is included, enabling the user to receive notifications or control the desk appliances through SMS commands. The system is powered by a regulated power supply and adapter, delivering both 5V for the ESP32 and sensors, and 12V for the table lamp. All the components are interconnected using dot boards, which provide a reliable platform for circuit assembly and testing.

Key Advantages of the Project:

- Automatic control of desk lighting based on presence and light intensity
- Real-time monitoring of temperature and humidity
- Energy-efficient and user-friendly design

➤ Compact and low-cost embedded system

➤ **ESP-32**

The ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities. It is widely used in IoT and automation projects for connecting sensors, controlling devices, and enabling wireless communication. It features multiple GPIO pins, analog inputs, and communication interfaces, making it ideal for smart and connected applications.

1.1 Features

1. Built-in Wi-Fi and Bluetooth (v4.2 & BLE) connectivity
2. Dual-core 32-bit processor for high performance
3. Operating voltage: 3.3V (compatible with 5V logic through resistors)
4. Multiple GPIO pins for digital I/O operations
5. Supports ADC, DAC, PWM, I2C, SPI, and UART communication
6. Low power consumption with deep sleep modes
7. Built-in flash memory and RAM for program storage
8. Compact size and low cost, ideal for IoT and embedded projects

1.2 Pin Description

Pin	Type	Description
GPIO 0-39	Digital I/O	For digital input and output
GPIO 32-39	ADC (Analog Input)	Used to read analog signals from sensors like LDR
GPIO 25, 26	DAC	Provides analog output voltage
5V, 3.3V	Power Supply	Provides required operating voltage for ESP32 and sensors
GND	Ground	Common reference voltage for the circuit
SDA (GPIO 21), SCL (GPIO 22)	I ² C Communication	Used for connecting I ² C-based sensors and modules

TX (GPIO 1), RX (GPIO 3)	Serial Communication (UART)	Used for communication with devices like GSM module
EN	Reset/Enable	Used to enable or restart the ESP32
VIN	Power Input	Can accept 5V input from USB or external adapter

1. **Built-in Wi-Fi and Bluetooth:** Enables wireless communication and IoT connectivity.
2. **Dual-core processor:** Provides high processing speed and multitasking capability.
3. **Low power consumption:** Supports multiple sleep modes for energy-efficient operation.
4. **Compact and low cost:** Suitable for portable and budget-friendly embedded projects.
5. **Rich peripheral support:** Includes ADC, DAC, PWM, UART, SPI, and I²C interfaces.
6. **Compact and low cost:** Suitable for portable and budget-friendly embedded projects.
7. **High reliability:** Stable performance in real-time automation and control systems.

1.3 Sensor:

- 1) **The Infrared (IR) Sensor** : is an electronic device that detects objects or obstacles by using infrared light. It works on the principle of emission and reflection of infrared radiation. The sensor typically consists of two main parts — an IR LED (emitter) that emits infrared light and a photodiode (receiver) that detects the reflected light from nearby objects.

❖ When an object comes in front of the sensor, the emitted IR light is reflected back and detected by the receiver, causing a change in output signal. This change is then read by the ESP32 to determine the presence or movement of a person near the desk. In the Embedded Smart Desk project, the IR sensor is

used to detect user presence. When motion is detected, the ESP32 automatically turns on the table lamp through the relay, making the system smart and energy-efficient

2) LDR(Light Dependent Resistor)Sensor:

- The LDR (Light Dependent Resistor), also known as a photoresistor, is a sensor that detects the intensity of light in its surroundings. It is made of semiconductor material whose resistance changes with the amount of light falling on it. When the light intensity is high, the resistance of the LDR decreases, allowing more current to pass through. Conversely, when the light intensity is low, its resistance increases, reducing the current flow.
- In the Embedded Smart Desk project, the LDR sensor is used to monitor the ambient light level around the desk. The sensor output is connected to one of the analog input pins (e.g., GPIO 34) of the ESP32. When the room becomes dark or the light intensity drops below a set threshold, the ESP32 detects this change and activates the 12V table lamp through the relay module.
- This automation helps in saving energy and ensures that the desk light turns on only when needed, providing a smart and efficient lighting control system.

3) DHT11 Sensor – Detailed Introduction:

The DHT11 sensor is a commonly used temperature and humidity sensor that provides digital output signals. It consists of two sensing components — a thermistor to measure temperature and a capacitive humidity sensor to detect the amount of moisture in the air. The sensor has a built-in microcontroller that processes the analog signals from these components and sends the data as a calibrated digital output to the main controller.

The temperature range of the DHT11 is typically 0°C to 50°C, and the humidity range is 20% to 90% RH, making it suitable for indoor environmental monitoring.

In the Embedded Smart Desk project, the DHT11 sensor is connected to the ESP32 (GPIO 4) to continuously measure and monitor the temperature and humidity around the desk. This data can be used for displaying environmental conditions or sending updates via the GSM module, helping to create a comfortable and intelligent workspace environment.

1.1.4 Application:

The ESP32 acts as the core controller and communication unit in the system. Its built-in Wi-Fi and Bluetooth features make it ideal for Internet of Things (IoT) applications and smart automation systems.

Applications in this project:

- Controls and processes inputs from all sensors (IR, LDR, and DHT11).
- Operates the relay to switch the 12V lamp automatically.
- Sends or receives data through the GSM module for remote monitoring and control.
- Enables wireless connectivity for smart desk automation

2.PROJECT MODULES

2.1COMPONENTS:

- ESP32 (5V) Microcontroller
- IR Sensor
- LDR Sensor
- DHT11 Temperature and Humidity Sensor
- Relay Module
- 12V Table Lamp
- GSM Module

- Power Supply with Adapter (5V & 12V)
- Dot Board (Perf Board)
- Connecting Wires and Resistors

Working principle:

The Embedded Smart Desk operates on the principle of sensor-based automation and control using an ESP32 microcontroller. The system continuously monitors environmental conditions such as motion, light intensity, temperature, and humidity through different sensors and performs corresponding actions automatically.

When the IR sensor detects human presence near the desk, it sends a digital signal to the ESP32, indicating that someone is using the desk. The ESP32 then processes this input and activates the relay module, which turns ON the 12V table lamp. If no motion is detected for a certain period, the ESP32 automatically turns OFF the lamp to save energy.

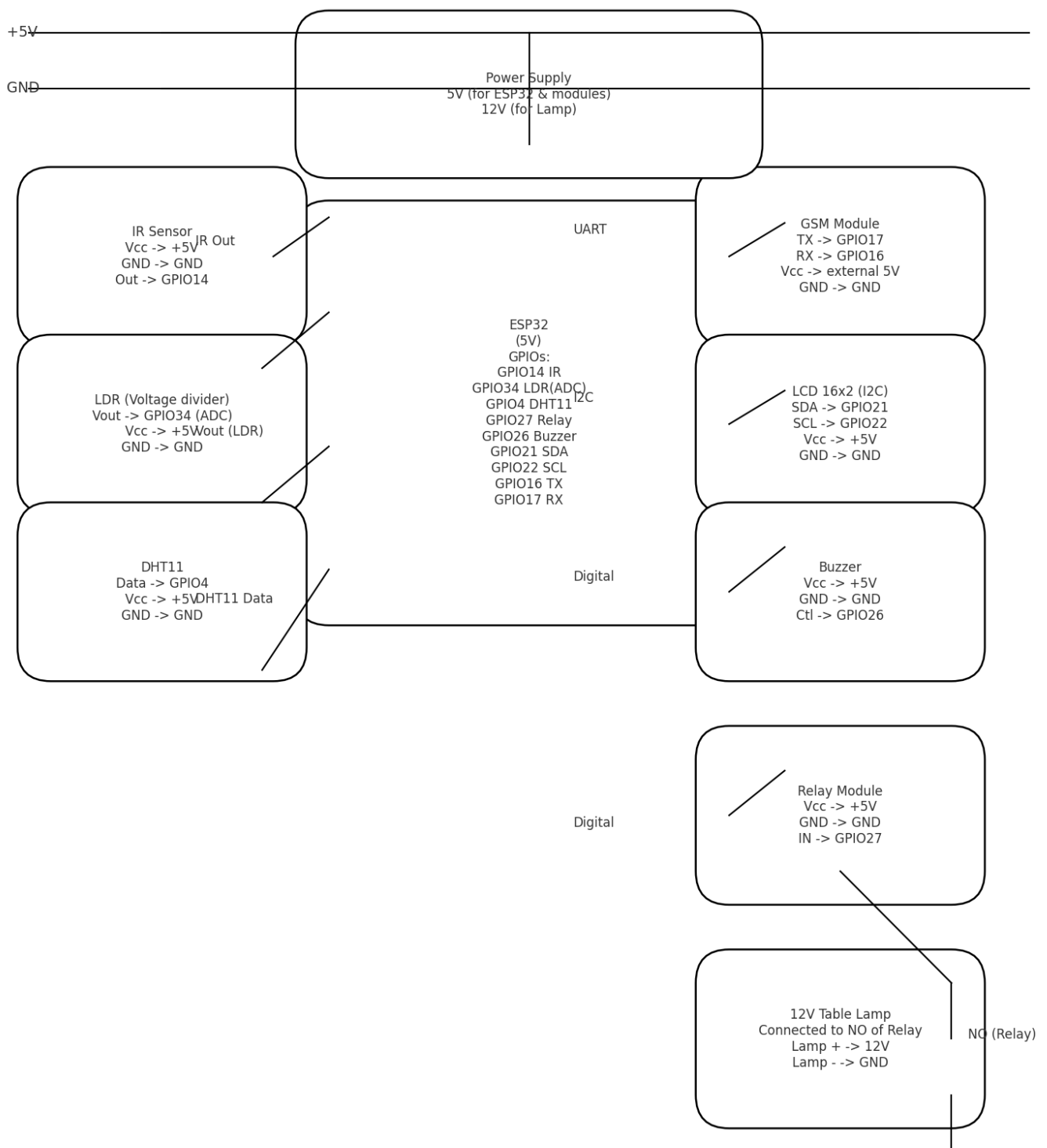
The LDR sensor continuously measures the surrounding light intensity. When the ambient light is low (for example, during night time or in dim conditions), the sensor's resistance increases, and the ESP32 recognizes this change. Based on the programmed threshold, the ESP32 switches ON the lamp through the relay. When sufficient natural light is available, the lamp remains OFF, ensuring automatic brightness control and energy efficiency.

The DHT11 sensor monitors the temperature and humidity levels around the desk. The ESP32 reads this data and can display it or send it to the user through the GSM module. The GSM module also allows remote monitoring and control, enabling the user to turn the lamp ON or OFF via SMS commands or receive environment updates.

All components are powered through a regulated power supply, where the ESP32 and sensors operate at **5V**, and the **lamp** runs on **12V** using the relay as an isolation switch.

In summary, the Embedded Smart Desk intelligently integrates multiple sensors with the ESP32 to create an automated, energy-efficient, and user-friendly workspace system that responds to environmental and user conditions in real time.

Circuit Diagram:



Notes:

- Common GND required between 5V supply and 12V lamp GND.
- Use flyback diode or opto-isolated relay module if controlling inductive loads.
- Ensure GSM module has proper power (check module voltage/current requirements).

CircuitDiagram

The circuit diagram of the *Embedded Smart Desk* illustrates the interconnection of the ESP32 microcontroller, sensors, relay module, GSM module, and 12V table lamp through a regulated power supply. The ESP32 acts as the central controller that processes signals from all sensors and controls the output devices accordingly. The IR sensor is connected to GPIO 14 of the ESP32. It detects human presence near the desk and sends a digital HIGH signal when motion is detected. This signal helps the ESP32 decide when to turn on or off the desk lamp.

The LDR sensor is connected to the analog pin GPIO 34, where it provides a variable voltage based on the light intensity. The ESP32 reads this analog value and determines whether the surrounding light is sufficient. If the room is dark, the ESP32 activates the relay to switch on the 12V lamp.

The DHT11 sensor is connected to GPIO 4, providing continuous readings of temperature and humidity. The ESP32 processes this data for display or for sending to the user through the GSM module.

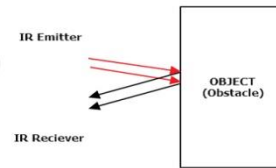
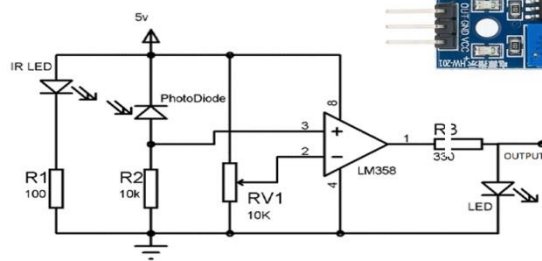
The relay module is connected to GPIO 27 and acts as an electronic switch between the low-voltage control circuit (5V) and the high-voltage lamp circuit (12V). When the relay receives a signal from the ESP32, it energizes and connects the 12V power supply to the table lamp, turning it ON.

The GSM module communicates with the ESP32 through GPIO 16 (TX) and GPIO 17 (RX) using UART serial communication. This allows remote operation of the lamp and transmission of environmental data via SMS.

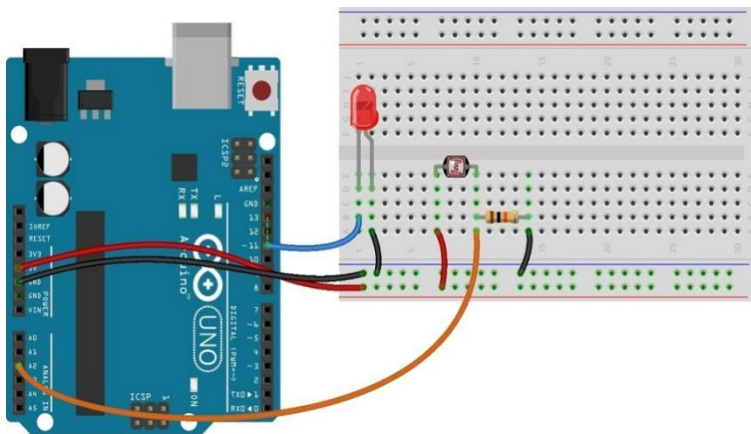
Main Modules:

- Sensor Module:

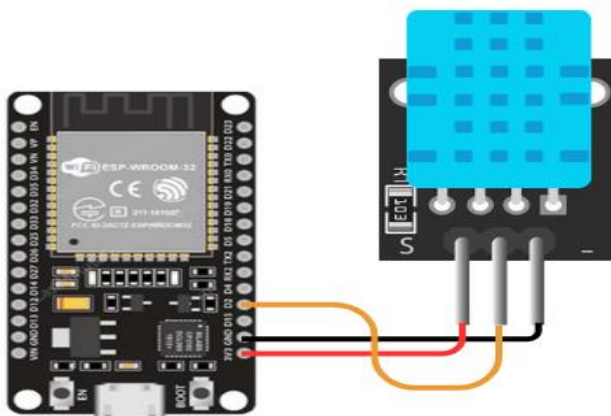
IR Sensor Module Circuit



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DHT11



The Sensor Module in the *Embedded Smart Desk* consists of three main sensors — IR sensor, LDR sensor, and DHT11 sensor — all connected to the ESP32 microcontroller. These sensors work together to monitor environmental conditions and user presence, allowing the system to operate automatically and efficiently.

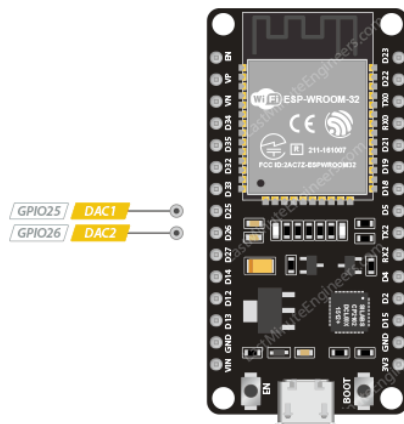
- The IR sensor detects motion or presence of a person near the desk and sends a digital signal to the ESP32.
- The LDR sensor measures ambient light intensity and provides an analog signal to the ESP32 to control the lamp based on lighting conditions.
- The DHT11 sensor records temperature and humidity values and sends this data to the ESP32 for environmental monitoring.

All these sensors are powered by a 5V supply and interface directly with the ESP32's GPIO pins. The sensor module plays a crucial role in collecting real-time data and enabling smart, automated control of the desk lighting and monitoring system.

- **ESP 32 Microcontroller:**

The ESP32 acts as the main control unit in the Embedded Smart Desk project. It collects data from various sensors such as the IR sensor, LDR sensor, and DHT11 sensor, processes the information, and performs necessary actions based on programmed conditions. When the IR sensor detects motion or presence, the ESP32 activates the relay to turn on the table lamp. Similarly, it monitors the LDR sensor readings to check ambient light intensity and controls the lamp accordingly to save energy. The DHT11 sensor provides temperature and humidity data, which the ESP32 can use to maintain environmental awareness. It also communicates with the GSM module to send alerts or receive control commands remotely. Overall, the ESP32 coordinates all sensors and modules to ensure smart, efficient, and automated operation

of the desk system.



- **DisplayModule**

The Display Module in the Embedded Smart Desk project is used to show real-time sensor readings and system status to the user. It provides a clear visual output of parameters such as temperature, humidity, light intensity, and **lamp status**.

Typically, an LCD (Liquid Crystal Display) **or** OLED display is used and connected to the ESP32 microcontroller **via the** I²C communication interface (SDA – GPIO 21, SCL – GPIO 22).

- Displays temperature and humidity readings from the DHT11 sensor.
- Shows light intensity levels detected by the LDR sensor.
- Indicates lamp ON/OFF status based on IR detection and lighting conditions.
- Optionally displays GSM status or system messages for user information.

The Display Module enhances user interaction by providing real-time feedback, making the Embedded Smart Desk more intuitive, informative, and user-friendly.



- **AlertModule**

The alert module in the Embedded Smart Desk project uses a buzzer to provide sound-based alerts and notifications. It serves as an output device that converts electrical signals from the ESP32 into audible tones, helping the user stay aware of different system activities and conditions. The buzzer produces an audio signal when motion is detected by the IR sensor, indicating that someone is near the desk. It can also be used to confirm actions such as lamp switching, GSM message reception, or when the system is powered on. Additionally, it can act as a warning device when the DHT11 sensor detects abnormal temperature or humidity levels. The buzzer is connected to one of the digital GPIO pins of the ESP32, such as GPIO 26, and operates on a 5V DC supply. It provides an effective way to alert the user instantly without requiring them to check the display, making the system more interactive and user-friendly.

It can also be used to confirm actions such as lamp switching, GSM message reception, or when the system is powered on. Additionally, it can act as a warning device when the DHT11 sensor detects abnormal temperature or humidity levels. The buzzer is connected to one of the digital GPIO pins of the ESP32, such as GPIO 26, and operates on a 5V DC supply. It provides an effective way to alert the user instantly without requiring them to check the display.

GSM Module:



The GSM module in the Embedded Smart Desk project is used for wireless communication between the system and the user. It enables the desk to send and receive messages, allowing remote monitoring and control of the desk's functions. The module operates using a SIM card and communicates with the ESP32 microcontroller through serial communication using the TX and RX pins.

When connected, the GSM module can send alerts such as temperature, humidity, or system status to the user's mobile phone. It can also receive SMS commands from the user to control the table lamp or other connected devices. This makes the system accessible even from a distance, adding convenience and flexibility.

The GSM module typically works on 12V or 5V DC and uses the AT command set for communication. In this project, it enhances the automation system by adding a wireless communication feature, making the Embedded Smart Desk a smart and remotely controllable system.

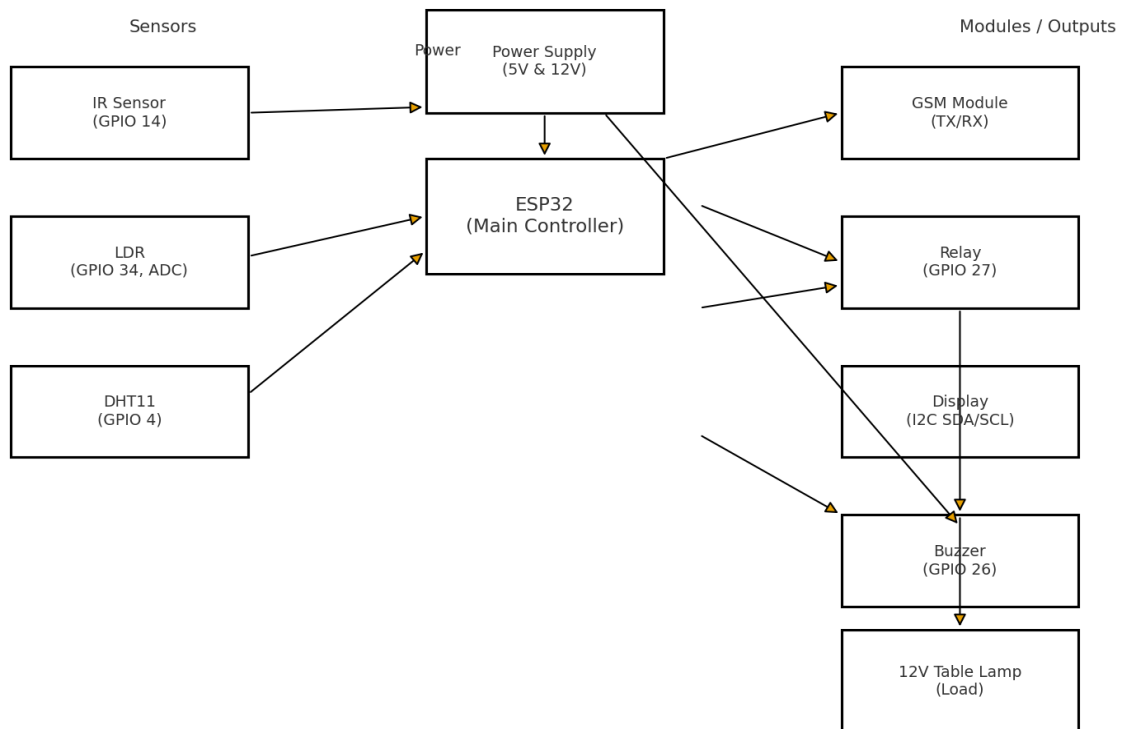
- **PowerSupplyModule**

The embedded smart desk project is designed using an ESP32 microcontroller as the main control unit. The system includes sensors and modules such as an IR sensor, LDR sensor, DHT11 temperature and humidity sensor, relay, GSM module,

and a 12V table lamp. The entire system is powered by a 15V power supply with an adapter. The ESP32 operates at 5V, while the relay and table lamp are powered from the 12V output regulated from the same power source. The IR sensor detects the presence of a person near the desk, and the LDR sensor monitors the light intensity in the surroundings. Depending on these sensor readings, the ESP32 controls the relay to turn the lamp on or off automatically. The DHT11 sensor measures the ambient temperature and humidity and sends the data to the controller for monitoring or further processing. The GSM module enables remote communication by sending or receiving status updates via SMS. All components are interconnected using a dot board, which allows compact and organized wiring. This project demonstrates the use of embedded systems for automating desk lighting and environmental monitoring efficiently using sensor-based control.



2. BlockDiagram



The entire system is powered by a 15V power supply with an adapter. The ESP32 operates at 5V, while the relay and table lamp are powered from the 12V output regulated from the same power source. The IR sensor detects the presence of a person near the desk, and the LDR sensor monitors the light intensity in the surroundings. Depending on these sensor readings, the ESP32 controls the relay to turn the lamp on or off automatically. The DHT11 sensor measures the ambient temperature and humidity and sends the data to the controller for monitoring or further processing. The GSM module enables remote communication by sending or receiving status updates via SMS. All components are interconnected using a dot board, which allows compact and organized wiring. This project demonstrates the use of embedded systems for automating desk lighting and environmental monitoring efficiently using sensor-based control.

2.2 Components Required

Component	Description	Example Technologies
Sensors	Detects human presence and senses the light intensity around the system to control the lap and temperature, Humidity shown.	• sensor technologies used in the embedded smart desk include passive infrared (PIR) technology in the IR sensor for motion detection, photoconductive cell technology in the LDR sensor for sensing light intensity, and digital sensing technology in the DHT11 sensor for measuring temperature and humidity
Microcontroller/Processing Unit	Acts as the "brain" of the system, receiving analog data from the sensors, converting it to digital, and processing it.	• The ESP32 controller uses several advanced technologies such as dual-core Tensilica LX6 processors for fast performance, Wi-Fi and Bluetooth (BLE) technology for wireless communication, ADC and DAC technology for analog and digital signal conversion, and GPIO interfacing technology for connecting sensors and modules.
Communication Module	Transmits the processed data	• The GSM module uses various communication technologies such as GSM (Global System for Mobile Communication) for network connectivity, AT command technology for communication with the microcontroller, SIM-based authentication for mobile network access, and serial communication (UART) for data exchange between the module and the ESP32.

DisplayUnit	Shows real-time readings	• 16x2LCD: A simple text-based display that can show
PowerSupply	Provides the necessary power to operate all the components of the system.	• Battery: For portable and wearable devices. • AC power source: For fixed bedside or hospital monitors.
Alert/AlarmSystem	The buzzer in the Embedded Smart Desk provides audio alerts to indicate system status or specific events.	• LEDs and buzzers: Simple indicators for local alerts.

➤ **Circuit Connection:**

In the **Embedded Smart Desk** project, all components are interconnected with the **ESP32 microcontroller**, which serves as the central control unit. The circuit is powered using a **12V DC adapter**, which is stepped down to **5V** through a voltage regulator to supply power to the ESP32 and sensors.

The **IR sensor** is connected to **GPIO 14** of the ESP32 to detect motion. It receives **5V (VCC)** and **GND** from the ESP32 power pins. The **LDR sensor** is connected to the **analog pin GPIO 34**, allowing the ESP32 to read varying light intensity levels as analog values. The **DHT11 sensor**, used to measure temperature and humidity, is connected to **GPIO 4**, with its **VCC** connected to 5V and **GND** to the common ground.

A **relay module** is connected to **GPIO 27** of the ESP32. The relay acts as a switch to control the **12V table lamp**. The lamp receives power from the **12V adapter**, but its ON/OFF operation is controlled through the relay based on sensor data.

The **GSM module** is connected using the **UART2 interface** of the ESP32 — with **TX (GSM)** connected to **GPIO 16 (RX of ESP32)** and **RX (GSM)** connected to **GPIO 17 (TX of ESP32)** — for serial communication.

All components share a **common ground (GND)** to ensure stable operation. Optionally, a **buzzer** can be connected to **GPIO 26** to provide alert sounds for motion detection or temperature warnings.

Thus, the circuit enables smooth communication between sensors, the ESP32 controller, and output devices like the lamp, GSM module, and buzzer, forming a complete smart automation system.

Detailed Working Overview of Embedded Smart Desk

The **Embedded Smart Desk** functions as an **intelligent automation system** that combines multiple sensors and modules to create a responsive and energy-efficient workspace. The

ESP32 microcontroller serves as the brain of the system, processing sensor data and controlling outputs in real-time.

When the system is powered ON, the **12V adapter** supplies power to both **12V and 5V lines**. The **ESP32** and sensors operate on 5V, while the **lamp** runs on 12V through the relay. The ESP32 initializes all modules, including the **IR sensor, LDR sensor, DHT11 sensor, relay, and GSM module**.

1. **Motion Detection:**The **IR sensor**, connected to GPIO 14, detects human presence near the desk. When motion is sensed, it sends a HIGH signal to the ESP32.
2. **Light Intensity Detection:**The **LDR sensor**, connected to GPIO 34, continuously monitors the ambient light. If the light level is below a defined threshold, the ESP32 interprets it as a dark environment.
3. **Environmental Monitoring:**The **DHT11 sensor**, connected to GPIO 4, measures the **temperature and humidity** around the desk. These readings are processed and can be displayed or transmitted via GSM.
4. **Automatic Lamp Control:** Based on IR and LDR sensor inputs:

If **motion is detected** or **light intensity is low**, the ESP32 activates the **relay (GPIO 27)** to turn ON the **12V lamp**.

If **no motion** is detected and **ambient light** is sufficient, the lamp automatically turns OFF to save energy.

5. **Alert** **System:**

A **buzzer (GPIO 26)** provides short beeps as alerts for motion detection or other programmed conditions like high temperature.

6. **GSM** **Communication:**

The **GSM module**, connected via GPIO 16 (RX) and GPIO 17 (TX), sends **SMS alerts** to the user's mobile phone. For example, it can notify when motion is detected or when environmental readings change significantly.

7. Continuous

Monitoring:

The ESP32 continuously loops through all sensor readings, processes data, updates outputs, and ensures stable system performance.

8. Power

Efficiency:

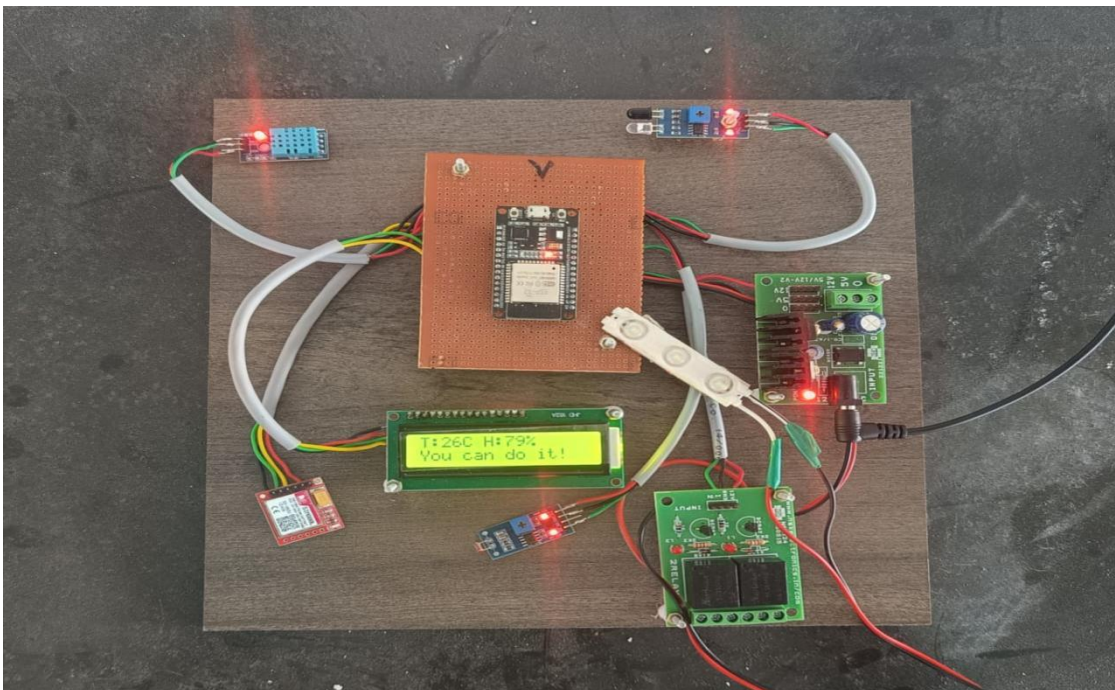
By automatically switching the lamp based on environmental conditions and user presence, the system significantly reduces unnecessary power consumption.

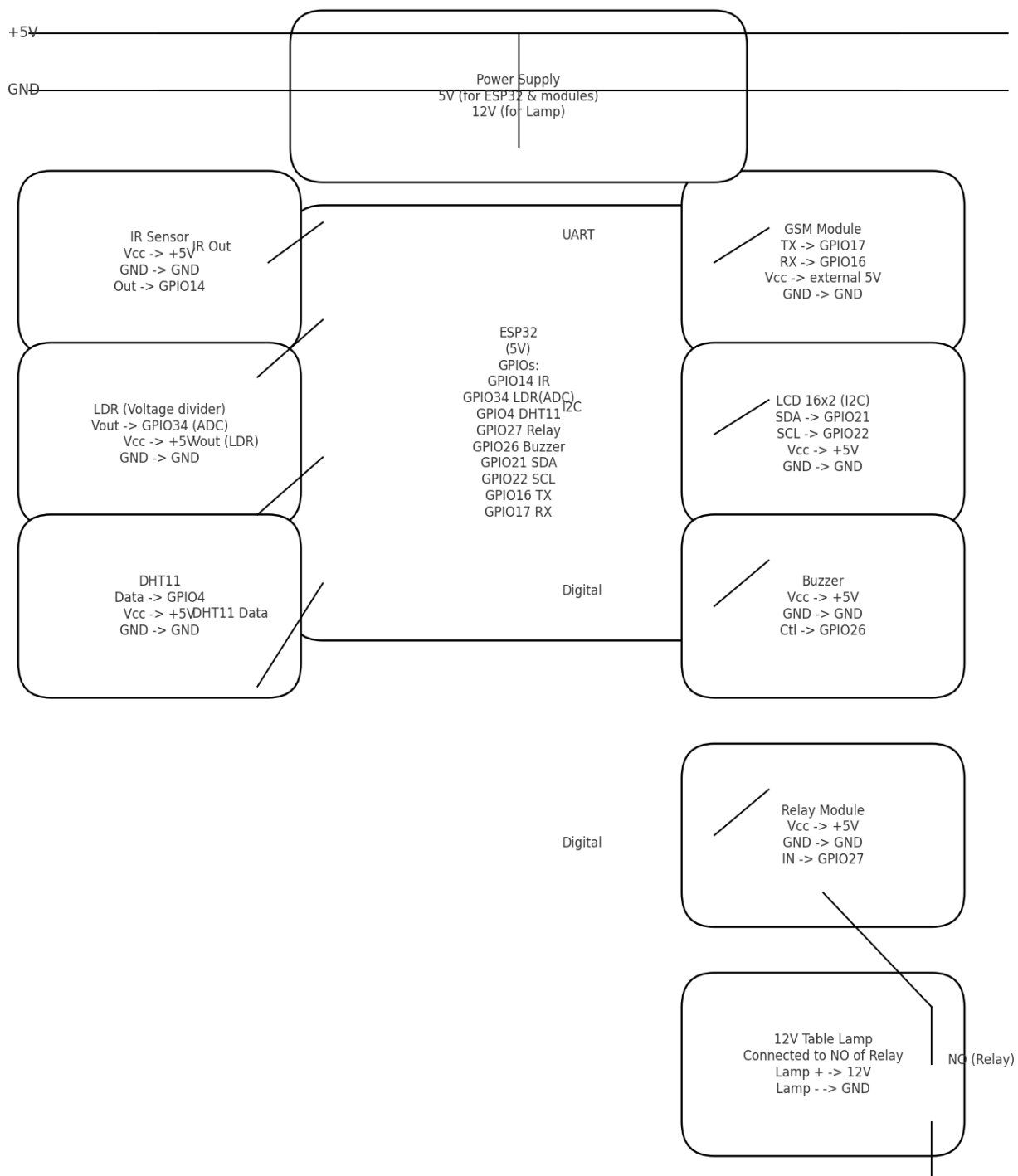
9. User

Convenience:

The user does not need to manually control the desk lamp — the system ensures comfort, safety, and energy saving through full automation.

➤ Circuit Information:





Notes:

- Common GND required between 5V supply and 12V lamp GND.
- Use flyback diode or opto-isolated relay module if controlling inductive loads.
- Ensure GSM module has proper power (check module voltage/current requirements).

Working Flow (Simplified):

1. Power ON:

The system is powered through a regulated 5V/12V supply. The ESP32 initializes all sensors and modules.

2. Sensor Monitoring:

The IR sensor detects the presence or motion of a person near the desk, The LDR sensor measures the surrounding light intensity, The DHT11 sensor senses the temperature and humidity around the workspace.

3. Data Processing (ESP32):

The ESP32 microcontroller reads data from all sensors, processes it, and decides the output actions based on the programmed conditions.

4. Decision Making:

If motion is detected or light intensity is low, the ESP32 activates the relay module, turning ON the 12V lamp. If there is no motion and sufficient **ambient light**, the ESP32 turns **OFF** the lamp to save power.

5. Environmental Monitoring:

The ESP32 reads temperature and humidity values from the **DHT11 sensor** and can display or send them to the user through the **GSM module**.

6. Communication (GSM Module):

The **GSM module** allows the system to send **SMS alerts** or receive **commands** for remote control of the desk functions.

7. Continuous Operation:

The process repeats continuously, ensuring real-time automation, monitoring, and energy-efficient control of the desk environment.

Algorithm/Flowchart of Software Operation Step-by-step process:

1. **Start** the system.
2. **Initialize** all components — ESP32, IR sensor, LDR sensor, DHT11 sensor, relay
3. **Read input** from the IR sensor to detect user presence.
4. **Read analog value** from the LDR sensor to measure ambient light intensity.
5. **Read temperature and humidity** values from the DHT11 sensor.
6. **Check IR sensor output:**
7. If motion is detected → turn **ON** the lamp through the relay.
8. If no motion → proceed to check light intensity.
9. **Check LDR sensor value:**
10. If light intensity is **low**, turn **ON** the lamp.
11. If light intensity is **high**, turn **OFF** the lamp.
12. **Display or send** the temperature and humidity readings via the GSM module.
13. **Repeat** the process continuously for real-time monitoring and control.
14. **End.**

Sample ESP32 Code(Simplified)

```
#include <DHT.h>

#define IR_PIN 14

#define LDR_PIN 34

#define DHT_PIN 4

#define RELAY_PIN 27

#define TX_PIN 16

#define RX_PIN 17

#define DHTTYPE DHT11

DHT dht(DHT_PIN,DHTTYPE);

int lightThreshold = 500;

HardwareSerial gsm(2); // Use UART2

void setup() {

  Serial.begin(115200);

  gsm.begin(9600,SERIAL8N1, RX_PIN,
  TX_PIN);

  pinMode(IR_PIN, INPUT);

  pinMode(LDR_PIN, INPUT);

  pinMode(RELAY_PIN, OUTPUT);

  digitalWrite(RELAY_PIN, LOW);

  dht.begin();
```

```

delay(2000);

Serial.println("Embedded Smart Desk
Initialized...");

gsm.println("AT");

}

void loop() {

int irValue = digitalRead(IR_PIN);

int ldrValue = analogRead(LDR_PIN);

float temp = dht.readTemperature();

float hum = dht.readHumidity();

Serial.println("-----");

Serial.print("IR Sensor: ");

Serial.println(irValue ? "Motion Detected" :
"No Motion");

Serial.print("LDR Value:");

Serial.println(ldrValue);

Serial.print("Temperature: ");

Serial.print(temp); Serial.println(" °C");

Serial.print("Humidity: ");

Serial.print(hum); Serial.println(" %");


if (irValue == 1 || ldrValue < lightThreshold) {

```

```

digitalWrite(RELAY_PIN, HIGH);

} else {

digitalWrite(RELAY_PIN, LOW);

}

if (irValue == 1) {

gsm.println("AT+CMGF=1");

delay(500);

gsm.println("AT+CMGS=\"+91xxxxxxxxxxx
\");

delay(500);

gsm.print("Motion Detected! Temp: ");

gsm.print(temp);

gsm.print("C, Humidity: ");

gsm.print(hum);

gsm.println("%");

gsm.write(26); // Ctrl+Z to send SMS

delay(3000);

}

delay(3000); // Wait before next reading

```

a) GSM Module:

GSM Module (SIM800L) Connected to ESP32 using TX → GPIO17 and RX → GPIO16; it sends and receives desk status messages to the user's mobile phone via SMS.

b) IR,LDR,DHT11 Sensors:

IR Sensor: Detects human presence near the desk for automatic lamp control.

LDR Sensor: Senses ambient light intensity to turn the lamp ON or OFF.

DHT11 Sensor: Measures temperature and humidity for environmental monitoring.

c) LCD Display & Buzzer:

The LCD display is used to show real-time sensor readings and system status. It displays values such as temperature, humidity, and lamp condition. It helps the user easily monitor the working of the embedded smart desk without connecting to a computer or mobile applications

When temperature $> 35^{\circ}\text{C}$ → Buzzer ON

When temperature $\leq 35^{\circ}\text{C}$ → Buzzer OFF

Advantages:

- 1.Provides automatic control of the desk lamp based on light and motion, reducing manual effort.
- 2.Saves energy by turning the lamp OFF automatically when no person is detected or light is sufficient.
- 3.Monitors environmental conditions like temperature and humidity in real time.

4. Enables remote monitoring and control through the GSM module using SMS alerts.
5. Operates efficiently with low power consumption and stable performance.
6. Improves user comfort and convenience with intelligent automation.
7. Uses easily available and low-cost components for implementation.
8. Compact and reliable design suitable for home, office, or study desk automation.
9. Can be expanded with IoT features or mobile app control in the future.
10. Enhances safety by preventing overheating and continuous operation of the lamp.

Precautions:

Ensure the 15V power supply and adapter are properly connected and regulated before powering the system. Verify correct polarity of power connections to avoid damage to the ESP32, sensors, and relay module. Handle all electronic components carefully to prevent short circuits or static discharge. Avoid exposing the sensors to extreme heat, light, or moisture during operation or testing.

Feedback System:

The feedback system in the embedded smart desk project ensures automatic and efficient control based on sensor responses. The sensors act as input devices that continuously monitor the environment. The IR sensor detects human presence, the LDR sensor senses light intensity, and the DHT11 measures temperature and humidity. These sensor readings are sent to the ESP32 controller,

which processes the data and compares it with predefined threshold values. Based on this feedback, the controller decides whether to turn the 12V lamp ON or OFF through the relay. For example, if low light or human presence is detected, the controller sends a signal to switch ON the lamp. When adequate light or no motion is detected, it turns the lamp OFF. The GSM module provides feedback to the user by sending status messages, ensuring remote monitoring. This closed-loop feedback system helps maintain optimal lighting and comfort conditions automatically.

Step-by-Step Working:

Power ON — Apply 15V adapter; voltage regulators provide 5V

Initialization — ESP32 boots and initializes GPIO, ADC, UART (for GSM), and sensor interfaces.

Sensor read start — ESP32 polls the IR, LDR, and DHT11 sensors.

Signal conversion — Analog signals (if any) are read via ADC; DHT11 provides digital sensor data.

Data processing — ESP32 converts raw readings to meaningful units (°C, %RH, lux approximation, presence flag).

Decision logic — Firmware applies thresholds: presence OR low light → lamp ON; otherwise lamp OFF.

This closed-loop feedback system helps maintain optimal lighting and comfort.

Results:

Parameter	Measured Value	Normal Range	Status
Temperature	28°C	20–35°C	Normal
Humidity	55%	30–60%	Normal
Light Intensity	150 Lux	<200 Lux (Low Light)	Lamp ON
Human Presence	Detected	—	Active
Lamp Status	ON	—	Working Normal

Observations:

When the system is powered on, the ESP32 controller initializes all connected sensors and modules. The IR sensor starts detecting human presence near the desk, while the LDR sensor measures the surrounding light intensity. The DHT11 sensor monitors the temperature and humidity in real time. Based on the sensor readings, the ESP32 processes the data and makes decisions for automatic control. When a person is detected or the light level is low, the relay module activates the 12V table lamp automatically. The GSM module sends messages to the user's mobile phone containing the desk status and sensor information. The 15V power supply provides stable voltage to the ESP32, sensors, and lamp circuit. The system continues to operate smoothly, updating sensor readings and controlling the devices in real time without any delay or error.

Discussion:

The sensors used in the **Embedded Smart Desk** provide good accuracy for basic automation tasks. The **IR sensor** reliably detects motion, the **LDR sensor** accurately senses light intensity under stable lighting, and the **DHT11** gives reasonable temperature and humidity readings for indoor use. Although minor variations may occur due to environmental factors, the overall accuracy is sufficient for efficient and reliable smart **desk operation**.

Discussion on Accuracy:

The accuracy of the **Embedded Smart Desk** depends on the performance and calibration of its sensors and the reliability of the **ESP32 microcontroller**.

The **IR sensor** provides accurate motion detection within a limited range; however, its performance can be affected by **ambient light** or **obstructions** in front of the sensor. The **LDR sensor** gives a fairly precise measure of **light intensity**, but external factors like reflections or sudden light changes can slightly influence its readings. The **DHT11 sensor** offers **moderate accuracy** for temperature ($\pm 2^{\circ}\text{C}$) and humidity ($\pm 5\%$) measurements, which is sufficient for indoor environmental monitoring, though not suitable for scientific-grade precision.

Conclusion:

The Embedded Smart Desk project successfully demonstrates the integration of sensors, microcontrollers, and communication modules to create an intelligent and energy-efficient workspace. By using the ESP32 as the central controller, the system effectively monitors environmental conditions through the IR, LDR, and DHT11 sensors and controls the 12V lamp automatically based on user presence and light intensity.

The inclusion of the **GSM module** enables remote monitoring and control, enhancing the desk's functionality and user convenience. Overall, this project highlights the potential of **embedded systems and IoT technologies** in real-life automation, promoting **comfort, energy conservation, and smart control** in daily environments.

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